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Low-dimensional Control under Embodiment Constraints: From LQR to Reinforcement Learning

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Abstract

Your English abstract here (mandatory if written in English and recommended otherwise).

Zusammenfassung

Hier die deutsche Zusammenfassung einfügen (notwendig).

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Chapter 1

Introduction

Over the recent years, the field of robotics has seen rapid advancements in the capabilities of neural-network-based implementations. An increase in global compute power[CITE: TODO compute], fueled by the so-called "AI race" of LLMs, has allowed researchers around the world to create even more extensive model architectures for more complex use cases [CITE: TODO llm example]. Transferring these structuresto physical bodies requires more space and on-device compute, often the limiting factors of embodiment. To bridge this gap between powerful models and limited resources, the approach of cheap design has become more common. As mentioned by N. Ay[CITE: TODO ay geometric], the core idea is to maintain an "embodied universal approximator", while reducing the overall complexity of the model, which can be defined based on the specific case. The most straightforward approach is inducing the underlying physical dynamics of the embodiment into the architecture, which is often of lower dimensionality than the actual sensors and actuators. Recent papers focusing on algorithms for the LQR[CITE: TODO lqr], apply this approach by offering implementations for enforcing low-rank controllers, leading to lower complexity in relation to parameterization. Still, this low-dimensional approach has yet to be extended[CITE: TODO extension] to more complex embodiments, and is missing a crucial part required for implementation: Robustness. This leaves us with the question: *Is it possible to train a low-dimensional controller for an embodied agent, that is performant yet robust?*

As robustness is a wide topic, in our case, we use it when referring to sensor- and actuator noise, as well as parametric uncertainties of the underlying model and external disturbances. When talking about performance, we mean reducing cost as much as possible in relation to an optimal controller. In the end, we want to show in which cases low-dimensional controllers are a viable alternative, and can thus reduce costs for hardware implementations.

To answer our question, we will first explain the core concepts and collect existing approaches for the LQR setting in section 2 and 3. In section 4, we will compare these implementations using the double integrator problem for LQR with

respect to general performance, as well as robustness. Building on that, we will artificially inflate our inputs and outputs via matrix multiplication in section 5 and additionally introduce deep RL baseline models into the comparison. For section 6, we will completely move away from LQR, and extend the setting to a perception-based implementation. Here we will compare different models with varying parameter complexity in regard to robustness, applying the concepts we learned from our earlier experiments. Finally, we will conclude by analyzing whether we indeed achieved our goal of constructing a robust controller, and stating how our approach can be applied to more complex settings.

Chapter 2

Preliminaries

Before turning to the experiments, this chapter revisits the concepts that all later parts of this thesis build on. Section 2.1 introduces linear state-space models and their discretization, Section 2.2 shows them on the double integrator that serves as our running example throughout, and Sections 2.3 and 2.4 recall the stability notions and the linear quadratic regulator. Sections 2.5 and 2.6 define the perturbation models and the robustness metrics used to evaluate every controller we construct.

2.1 Linear State-Space Models

In control theory, the evolution of an agent and together with its environment is described by a *state-space model*: a rule that assigns to every point of a state space the instantaneous rate at which the system moves through it. Throughout this thesis we restrict attention to the class of *linear time-invariant* (LTI) systems, for which this rule is linear in both the state and the input and does not itself depend on time[CITE: ControlSystemsTheoryAndDesign.pdf].

A continuous-time LTI system is given by

$$\dot{x}(t) = A_c\, x(t) + B_c\, u(t), \qquad x(0) = x_0, \tag{2.1a}$$

$$y(t) = C\, x(t) + D\, u(t), \tag{2.1b}$$

where $x(t) \in \mathbb{R}^n$ is the *state vector*, $u(t) \in \mathbb{R}^m$ the *input vector* chosen by the agent, and $y(t) \in \mathbb{R}^q$ the *output vector*, i.e. the output observed and thus available to the feedback controller. Accordingly, $A_c \in \mathbb{R}^{n\times n}$ is called the *system matrix*, $B_c \in \mathbb{R}^{n\times m}$ the *input matrix*, $C \in \mathbb{R}^{q\times n}$ the *output matrix*, and $D \in \mathbb{R}^{q\times m}$ the *feedthrough matrix*.

The two equations in (2.1) play fundamentally different roles, and it is helpful explaining their distinction. Equation (2.1a) is a differential equation: it does not

compute the state, which is assumed to be known at every instant, but prescribes its *rate of change*. The pair (A_c, B_c) therefore governs how the system *behaves*, with A_c describing its autonomous dynamics and B_c describing how the input of the agent influences them. Equation (2.1b), in contrast, contains no derivative at all. It is an instantaneous readout, and the pair (C, D) therefore describes how the system is *observed* rather than how it evolves. We set $D = 0$ throughout, which is the standard assumption for systems in which the actuator has no direct instantaneous influence on the sensor[CITE: either same notes or textbook].

Given an input signal $u(\cdot)$, the trajectory of (2.1a) is obtained by integration and is given in closed form by the variation-of-constants formula

$$x(t) = e^{A_c t} x_0 + \int_0^t e^{A_c (t-s)} B_c \, u(s) \, \mathrm{d}s, \tag{2.2}$$

where $e^{A_c t}$ denotes the matrix exponential[CITE: same textbook]. This makes the interpretation of A_c precise: in the scalar case $\dot{x} = ax$ yields $x(t) = e^{at} x_0$, so a is a growth *rate* rather than a value. Equation (2.2) generalises this, with the eigenvalues of A_c acting as growth rates along the corresponding eigendirections.

Full state measurement. Generally, we assume that the state is fully measurable and therefore set $C = I_n$, which reduces (2.1b) to $y(t) = x(t)$. However, when we introduce disturbances through noise, this will no longer be the case. More details on this will be given in Section 2.5. The distinction between $u = -Kx$ and $u = -Ky$ is therefore critical, since the former admits an optimal solution while the latter might not; we return to this point in Section 2.4.

2.1.1 Discretization

Digital controllers, as in our case, do not act continuously: they sample the output at a fixed rate and hold the computed controlinput constant until the next sampling point. The learning algorithms studied in this thesis operate on such sampled trajectories as well, so our model has to reflect this. The standard approach is the *zero-order hold* (ZOH)[CITE: find reference]. Fixing a sampling period $h > 0$ and writing $x_k \coloneqq x(kh)$, the input is assumed to be piecewise constant,

$$u(t) = u_k \qquad \text{for } t \in [kh,\, (k+1)h), \quad k \in \mathbb{N}_0. \tag{2.3}$$

Substituting (2.3) into (2.2) over a single sampling interval and applying the change of variables $\tau = (k+1)h - s$ yields the discrete-time model

$$x_{k+1} = A_d \, x_k + B_d \, u_k, \tag{2.4}$$

with

$$A_d = e^{A_c h}, \qquad B_d = \left(\int_0^h e^{A_c \tau} \, \mathrm{d}\tau \right) B_c. \tag{2.5}$$

Two clarifications need to be made. First, (2.4) is not an approximation: it reproduces the continuous-time trajectory exactly at the sampling instants, provided the input is genuinely piecewise constant. This distinguishes ZOH from other schemes such as the forward Euler discretization $A_d \approx I + hA_c$, $B_d \approx hB_c$, which is only first-order accurate.[1] Second, the left-hand side of (2.4) is the actual next state rather than a derivative. The identity $(x_{k+1} - x_k)/h \approx \dot{x}(kh)$ recovers the continuous picture, and the identity block contained in A_d but not in A_c is exactly the state carried over from the previous step.

Unless stated otherwise, all experiments in this thesis are carried out in discrete time with sampling period $h = 0.01$, and we drop the subscript d in the following, writing simply A and B.

2.2 The Double Integrator

To make the comparisons across all chapters meaningful, all experiments are built on a single system: the *double integrator*, a frictionless point mass of mass m moving along a line under an applied force u[CITE: find some reference]. Its dynamics are Newton's second law,

$$m\,\ddot{p}(t) = u(t), \tag{2.6}$$

where $p(t)$ denotes the position of the mass. We simplify by using $m = 1$, so that force and acceleration coincide numerically. Choosing position and velocity as the state, $x = [\,p,\ v\,]^\top \in \mathbb{R}^2$, the second-order equation (2.6) decomposes into the first-order system

$$\dot{p}(t) = v(t), \qquad \dot{v}(t) = u(t), \tag{2.7}$$

in which the first equation is the definition of velocity and the second is (2.6) itself. In the notation of (2.1) this gives

$$A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = I_2, \qquad D = 0. \tag{2.8}$$

The entry $(A_c)_{12} = 1$ should be read as stated in Section 2.1: the rate of change of position is given by the current velocity.

[1] For the double integrator of Section 2.2 the two approaches happen to agree on A_d but *not* on B_d, where forward Euler gives $[0,\ h]^\top$ instead of the correct $[h^2/2,\ h]^\top$.

Because $A_c^2 = 0$, the matrix exponential in (2.5) truncates after two terms, $e^{A_c h} = I + A_c h$, and the discretization is available in closed form:

$$A_d = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix}, \qquad B_d = \begin{bmatrix} h^2/2 \\ h \end{bmatrix}. \tag{2.9}$$

This system is the minimal nontrivial choice for our purposes. It is second-order, so the state is not directly actuated and the controller must account for momentum; it is controllable for every $h > 0$; and it is not open-loop stable, since A_c has a double eigenvalue at the origin and A_d correspondingly a double eigenvalue at 1. The last property matters for the learning experiments: an uncontrolled or badly initialised policy diverges, so the assumption of a known stabilising initial gain (Section 2.4) is a real restriction. These properties are made precise in Section 2.3.

2.3 Characteristics of a system

Now that we have defined our system, before we can optimize over it, it is important to clear up different characteristics of the same. The three we need are properties of the model and not of any particular controller. Stability is what the controller has to establish in the first place, controllability is what guarantees that it can, and observability is what allows a controller that only sees y to act as if it actually recorded x.

2.3.1 Stability

The first of these is stability, which guarantees that deviations in the system don't become too drastic, and thus cause some kind of failure. Mathematically, stability is a statement about a *solution* of the recursion $x_{k+1} = f(x_k, k)$, and is defined the following way[CITE: Astrom & Wittenmark, "Computer-Controlled Systems", p.77]:

Definition 2.1 (Stability)**.** Let $x^0(\cdot)$ be a solution of $x_{k+1} = f(x_k, k)$. The solution $x^0(\cdot)$ is called *stable* if for every $\varepsilon > 0$ there exists a $\delta(\varepsilon, k_0) > 0$ such that every solution $x(\cdot)$ with $\|x(k_0) - x^0(k_0)\| < \delta$ satisfies

$$\|x(k) - x^0(k)\| < \varepsilon \qquad \text{for all } k \geq k_0.$$

It is called *asymptotically stable* if it is stable and in addition $\|x(k) - x^0(k)\| \to 0$ as $k \to \infty$.

In the definition, ε bounds how far a perturbed trajectory is allowed to deviate, and δ says how small the initial distance must be to stay inside that bound. Asymptotic additionally requires the deviation to vanish with increasing time.

For linear systems this simplifies considerably. If $x(\cdot)$ and $x^0(\cdot)$ both solve $x_{k+1} = Ax_k$, then their difference solves the same recursion, so the behaviour of *one* solution decides the behaviour of all of them. Stability is thus a property of the system, explicitly the matrix A. This is stated as a theorem by[CITE: Astrom & Wittenmark, "Computer-Controlled Systems"

Theorem 2.1 (Asymptotic stability of discrete-time LTI systems)**.** *The discrete-time LTI system $x_{k+1} = Ax_k$ is asymptotically stable if and only if all eigenvalues of A lie strictly inside the unit disk, i.e.*

$$\rho(A) \coloneqq \max_i |\lambda_i(A)| < 1 \tag{2.10}$$

Our double integrator sits precisely on the boundary. From (2.9), A_d has the double eigenvalue $\lambda_{1,2} = 1$, so $\rho(A_d) = 1$ and Theorem 2.1 fails. It is not only non-asymptotically stable but actually unstable. We can see that in the matrix powers

$$A_d^k = \begin{bmatrix} 1 & kh \\ 0 & 1 \end{bmatrix}, \tag{2.11}$$

which are growing without bound in k. This is also logical from a physical point of view as a frictionless mass with a nonzero initial velocity continues its trajectory forever if nothing acts on it. Every controller we construct therefore has to *create* stability rather than preserve it.

2.3.2 Controllability

Whether a system is controllable or not indicates if, independently from the controller, it is possible to reach any state inside its state space in a given time. A more formal definition is given in [CITE: ControlSystemsTheoryAndDesign]:

Definition 2.2 (Controllability)**.** The system is said to be *controllable* if for any initial state x_0 and any final state x_f there exist a horizon N and inputs $u_0, \dots, u_{N-1}$ such that the solution of (2.4) satisfies $x_N = x_f$. Otherwise, the system is said to be *uncontrollable*.

To check whether a given system is actually controllable, we need to compute the *controllability matrix*[CITE: ControlSystemsTheoryAndDesign]

$$\mathcal{C}(A, B) = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \in \mathbb{R}^{n \times nm}, \tag{2.12}$$

and then verify that it has full rank n; the system is controllable if and only if it does[CITE: ControlSystemsTheoryAndDesign]. Note that (2.12) is stated for a general input matrix B; for a single input ($m = 1$) it reduces to the square matrix $[\, b \;\; Ab \;\; \cdots \;\; A^{n-1}b \,]$.

Applying this to our double integrator, we obtain

$$\mathcal{C}(A_d, B_d) = \begin{bmatrix} B_d & A_dB_d \end{bmatrix} = \begin{bmatrix} h^2/2 & 3h^2/2 \\ h & h \end{bmatrix}, \qquad \det \mathcal{C}(A_d, B_d) = -h^3, \tag{2.13}$$

which is nonzero for every sampling period $h > 0$. The system is therefore controllable at every sampling rate, as claimed in Section 2.2.

For Section 2.4 we will not need the strong Definition 2.2 but can rather suffice with the weaker notion of *stabilizability* [CITE: ControlSystemsTheoryAndDesign]:

Definition 2.3 (Stabilizability)**.** The pair (A, B) is called *stabilizable* if there exists a gain $K \in \mathbb{R}^{m\times n}$ such that $\rho(A - BK) < 1$, i.e. if the closed loop obtained from (2.4) with $u_k = -Kx_k$ can be made asymptotically stable. Equivalently, (A, B) is stabilizable if and only if every mode of A that is not controllable is already asymptotically stable.

However, to decide whether our system is stabilizable, we do not need to construct this specific K. The Popov-Belevitch-Hautus (PBH) test[CITE: TODO: Find fitting] allows us to employ it as a rank condition on the relevant eigenvalues:

Theorem 2.2 (PBH test for stabilizability)**.** *The pair (A, B) is stabilizable if and only if*

$$\operatorname{rank}\begin{bmatrix} A - \lambda I_n & B \end{bmatrix} = n \qquad \text{for all } \lambda \in \mathbb{C} \text{ with } |\lambda| \geq 1. \tag{2.14}$$

Comparing (2.14) with (2.12) shows that the latter applies the same rank condition as the former, however imposed for *every* $\lambda \in \mathbb{C}$. Thus, controllability implies stabilizability but not vice versa. The difference contains systems which have uncontrollable directions decaying on their own, and a controller can hence never influence.

As our double integrator is already controllable by (2.13), it is therefore also stabilizable.

2.3.3 Observability

Finally, we need to take a look at observability. Above we assumed that we can actually observe the state of the system we are modelling. However, it is much more realistic that we only collect a recording of it, for example through sensors, and need to estimate real state of the system. The notion of observability describes whether we can determine this state from our control input u and our recorded response y alone [CITE: ControlSystemsTheoryAndDesign]:

Definition 2.4 (Observability)**.** The system with state-space model (2.4) and output $y_k = Cx_k$ is said to be *observable* if the initial state x_0 can be uniquely determined from the time history of the input u_k and the output y_k over a finite interval. Otherwise, the system is said to be *unobservable*.

Dual to the controllability matrix, we can derive the *observability matrix*

$$\mathcal{O}(C, A) = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{pn \times n}, \tag{2.15}$$

and make use of the following theorem[CITE: ControlSystemsTheoryAndDesign]:

Theorem 2.3 (Observability rank condition)**.** *The system is observable if and only if the observability matrix $\mathcal{O}(C, A)$ has full rank n.*

Using Theorem 2.3 for our double integrator is actually quite simple: we set $C = I_2$ in (2.8), so $\mathcal{O}(C, A_d)$ already contains the identity in its first block rows and has full rank. The system is thus observable.

If we cannot uniquely determine x_k from our observation, we work with an estimate $\hat{x}_k$ instead. Running a copy of the model, $\hat{x}_{k+1} = A\hat{x}_k + Bu_k$, the estimation error $\tilde{x}_k \coloneqq x_k - \hat{x}_k$ follows the dynamics

$$\tilde{x}_{k+1} = A\,\tilde{x}_k, \qquad \tilde{x}_0 = x_0 - \hat{x}_0, \tag{2.16}$$

which is again just the system without control input. The decay of the error therefore depends only on the eigenvalues of A. If the system is unstable, the error grows infinitely[CITE: ControlSystemsTheoryAndDesign] and for our example it does so linearly, by (2.11). Observability fixes this by correcting the model with the measurement, $\hat{x}_{k+1} = A\hat{x}_k + Bu_k + L(y_k - C\hat{x}_k)$, which turns (2.16) into $\tilde{x}_{k+1} = (A - LC)\tilde{x}_k$. If the system is observable, we can choose a L such that the eigenvalues of $A - LC$ lie inside the unit disk, making it stable.

As with stabilizability, the weaker requirement is called *detectability*, and it is the condition we will rely on in Section 2.4.

Definition 2.5 (Detectability)**.** The pair (A, C) is called *detectable* if there exists an observer gain $L \in \mathbb{R}^{n \times p}$ such that $\rho(A - LC) < 1$. Equivalently, (A, C) is detectable if and only if every unobservable mode of A is already asymptotically stable.

Detectability is the dual of Definition 2.3 which gives us that (A, C) is detectable if and only if $(A^\top, C^\top)$ is stabilizable[CITE: ControlSystemsTheoryAndDesign]. This follows as $\rho(A - LC) = \rho\big((A^\top - C^\top L^\top)\big)$. Transposing the test for stabilizability (2.14) leaves us with the dual rank test:

Theorem 2.4 (PBH test for detectability)**.** *The pair (A, C) is detectable if and only if*

$$\operatorname{rank} \begin{bmatrix} A - \lambda I_n \\ C \end{bmatrix} = n \qquad \text{for all } \lambda \in \mathbb{C} \text{ with } |\lambda| \geq 1. \tag{2.17}$$

With $C = I_2$ the matrix in (2.17) contains an identity block and has full rank for every λ, so our system is observable and therefore detectable. The reason we introduce the weaker notion here is Theorem 2.5, whose hypothesis is the detectability of $(A, Q^{1/2})$ rather than (A, C). This shifts the focus from what the sensor sees, to which directions of the state space are penalised by the cost.

The topic of observability will become important again in Section 2.5, where we deliberately introduce noise so that $y_k \neq x_k$. This doesn't change our matrix $C = I_2$, so the rank condition of Theorem 2.3 still holds and the system remains observable by definition. However, y_k is no longer a deterministic function of x_k, so we can never estimate the state exactly. We revisit this and its implications there.

2.4 The Linear Quadratic Regulator

Now that we have defined our state-space model and its important characteristics, it is time to introduce our controller, which is based on the *linear quadratic regulator* (LQR) problem.

Here, the setting is that our system starts at an arbitrary initial state in its state space, and the goal of our controller is to return to the zero state as fast as possible. At the same time, the control input should be limited to a realistic quantity, such that it could actually be produced by an actuator. Both requirements can be combined into a single cost function, where Q describes the former, while R associates costs to control input. Over a finite horizon T, this gives the cost[CITE: OptimalAndRobustControl.pdf]

$$V_T(x_0, u) = \int_0^T \left(x(t)^\top Q\, x(t) + u(t)^\top R\, u(t) \right) \mathrm{d}t, \qquad Q = Q^\top \succeq 0, \quad R = R^\top \succ 0. \tag{2.18}$$

We notice to further restrictions here: Q is required to be positive semi-definite, meaning deviating from the origin isn't reducing costs, and R must be positive definite, such that every control input results in a cost.

As the original formula relates to a continuous problem with a fixed endtime T, it doesn't fit well to our goal of evaluating over an infinite discrete horizon. The following formula offers exactly that:

$$J(x_0, u) = \sum_{k=0}^{\infty} \left(x_k^\top Q\, x_k + u_k^\top R\, u_k \right). \tag{2.19}$$

This sum does not necessarily converge. If the closed loop does not decay, the costs do not vanish and J tends to $+\infty$. Using this as our definition, J becomes well defined for every gain.

2.4.1 The optimal gain

What makes the LQR problem special is that its solution is a static linear feedback law: the optimal input at time k depends only the current state of the model. Introducing a gain matrix $K \in \mathbb{R}^{m \times n}$, we write it as [2]

$$u_k = -Kx_k. \tag{2.20}$$

Our controller is thus reduced to choosing a matrix, and the optimal controller is given by the optimal gain matrix K^*.

As we are dealing with a discrete-time system, that gain is obtained from the *discrete algebraic Riccati equation* (DARE)

$$P = A^\top PA - A^\top PB \left(R + B^\top PB\right)^{-1} B^\top PA + Q, \tag{2.21}$$

whose stabilizing solution $P = P^\top \succeq 0$ yields

$$K^* = \left(R + B^\top PB\right)^{-1} B^\top PA, \qquad u_k^* = -K^* x_k. \tag{2.22}$$

The matrix P is additioanlly our *cost-to-go* matrix, meaning that we can use it to calculate the optimal cost incurred from any state x onwards with

$$J^*(x) = x^\top Px. \tag{2.23}$$

Whether (2.21) has such a solution at all is answered by the structural properties of Section 2.3[CITE: OptimalAndRobustControl.pdf]:

Theorem 2.5 (Existence and uniqueness)**.** *Let (A, B) be stabilizable and let $(A, Q^{1/2})$ be detectable. Then the DARE (2.21) has a unique symmetric positive semi-definite solution P, which is the only stabilizing one, and the resulting closed loop satisfies $\rho(A - BK^*) < 1$. If $(A, Q^{1/2})$ is observable, P is positive definite.*

2.4.2 The cost as a function of the gain

Equation (2.22) solves the problem, but it solves it directly and tells us nothing about the quality of non-optimal gains. The learning algorithms of the following chapters never see the model itself; they search over gains directly, and for that we need the cost as a function of K rather than of an input sequence.

Fix a gain K and write the resulting closed loop as $x_{k+1} = A_K x_k$ with $A_K \coloneqq A - BK$. Substituting (2.20) into (2.19) shows that the cost is again a quadratic

[2] Note that the minus in front of the K has been chosen as the convention for this paper. This turns our closed loop into the format $A - BK$, while some referenced literature uses $A + BK$. Formulas cited have been adapted to our convention by replacing K with $-K$.

form, $J(x_0, K) = x_0^\top P_K x_0$, where P_K now solves the far simpler discrete *Lyapunov equation*

$$P_K = A_K^\top P_K A_K + Q + K^\top R K, \tag{2.24}$$

which is linear in the unknown P_K and has a unique solution whenever $\rho(A_K) < 1$. This restricts us to the set of stabilizing gains

$$\mathcal{K} \coloneqq \left\{ K \in \mathbb{R}^{m \times n} : \rho(A - BK) < 1 \right\}, \tag{2.25}$$

outside of which we set the cost to $+\infty$, consistent with the remark after (2.19).

One dependency remains, namely on the initial state. Evaluating the gain based on a single x_0 would reward gains that happen to fit that one point, so we instead draw the initial state from a distribution, $x_0 \sim \mathcal{D}$, and average. Writing $\Sigma \coloneqq \mathbb{E}_{x_0 \sim \mathcal{D}}[x_0 x_0^\top]$ for its second moment, the objective becomes

$$f(K) \coloneqq \mathbb{E}_{x_0 \sim \mathcal{D}} \left[x_0^\top P_K x_0 \right] = \operatorname{tr}\left(P_K \Sigma \right), \tag{2.26}$$

using the cyclic property of the trace. Equation (2.26) is the object that every remaining chapter of this thesis minimizes, and the algorithms differ only in how much of the model is known to them.

Throughout this thesis we set $Q = I_2$ and $R = 1$, which weighs position, velocity and control effort equally. Initial states are drawn uniformly from the box $[-10, 10] \times [-1, 1]$.

2.5 Uncertainty and Disturbance Models

Until now, we assumed that we can fully observe the system, and that its behaviour is governed by exactly the dynamics we know. However, in reality there are influences on such systems which we don't clearly know of. Especially for embodied agents this is important, as the world is a messy place.

We will thus introduce four different disturbances to be included in the model, commonly known as noise. They are distinguished by the point at which they affect the loop: what the controller *observes* (measurement noise), what its output actually *does* (control noise), what the plant *is* (parametric uncertainty), and what else *acts* on it (external disturbance). Each of them can be activated separately, so that we can clearly isolate the cause of an effect. Throughout this section k indexes the sampling instants, and σ denotes the standard deviation of the respective noise.

2.5.1 Measurement Noise

This one mainly relates to observability: we influence the sensors used to measured the model state, such that capturing the same state twice would not lead to the same observation y_k. Let n_k^{m} denote the measurement noise at step k. Then

$$n_k^{\mathrm{m}} \sim \mathcal{N}\big(0, \sigma_{\mathrm{m}}^2 I_n\big), \qquad y_k = x_k + n_k^{\mathrm{m}}, \qquad \sigma_{\mathrm{m}} \in \mathbb{R}_+, \tag{2.27}$$

drawn independently at every step.

While one might think that this noise removes observability for us, it doesn't. The output matrix is still $C = I_2$, so the rank condition of Theorem 2.3 holds unchanged and observability is given.

However, it does remove the deterministic link between state and output, so we can't identify x_k exactly with any amount of measurements. Using our noisy observation in the feedback law (2.20) and $u_k = -Ky_k$, the closed loop becomes

$$x_{k+1} = (A - BK)\, x_k - BK\, n_k^{\mathrm{m}}. \tag{2.28}$$

So even a perfectly stabilizing K can no longer bring the state exactly to the origin, but rather to a distribution around it.

2.5.2 Control Noise

Control noise, as the name says, acts on the applied control input. With it, we try to model thedifference between the intended movement of an actuator, and its actual action due to different factors. Here we deal with *multiplicative* noise, which scales with the input. In detail this means that the applied control input $\tilde{u}_k$ is obtained as

$$n_k^{\mathrm{c}} \sim \mathcal{N}\big(0, \sigma_{\mathrm{c}}^2 I_m\big), \qquad \tilde{u}_k = (1 + n_k^{\mathrm{c}}) \odot u_k, \tag{2.29}$$

where $\odot$ denotes the elementwise product, so that σ_{c} is the *relative* error. Applying this to the input, the next step becomes

$$x_{k+1} = A\, x_k + B\, \tilde{u}_k. \tag{2.30}$$

2.5.3 Parametric Uncertainty

This noise acts directly on the matrices of the model. Here, we either decide to keep it time invariant, meaning $\Delta_k = \Delta_0$ for all k, or to resample it at every step. The former models a wrongly identified system, while the latter simulates one that continuously changes. As with measurement noise, it is additive. Applied, it acts on the system the following way:

$$\left(\Delta_k^A\right)_{ij},\ \left(\Delta_k^B\right)_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\left(0, \sigma_{\text{p}}^2\right), \qquad x_{k+1} = \left(A + \Delta_k^A\right) x_k + \left(B + \Delta_k^B\right) \tilde{u}_k. \quad (2.31)$$

Worth noting is that resampling every step leaves the model correct *on average*, since $\mathbb{E}[\Delta_k] = 0$, and mainly adds variance to the trajectory. This stands in contrast to the time invariant version, where we have an discrepancy between what we compute actions on, and what actually happens.

2.5.4 External Disturbance

This noise models external influences, like a constant wind pushing on our agent, or periodic disturbances, for example rain hitting it with some frequency. Depending on the type, the disturbance $n_k^{\text{d}} \in \mathbb{R}^n$ is drawn as

$$\begin{aligned}
&\text{Gaussian:} && n_k^{\text{d}} \sim \mathcal{N}\left(0, \sigma_{\text{d}}^2 I_n\right) \quad \text{independently at every step,} && (2.32a)\\
&\text{Sinusoidal:} && n_k^{\text{d}} = \sigma_{\text{d}} \sin(0.01\, k)\, \mathbf{1}_n, && (2.32b)\\
&\text{Constant:} && n_k^{\text{d}} = n_0^{\text{d}} \sim \mathcal{N}\left(0, \sigma_{\text{d}}^2 I_n\right) \quad \text{for all } k. && (2.32c)
\end{aligned}$$

Only the first one can really be considered as noise; the other are more deterministic values, where σ_{d} acts as the amplitude of it. The sinusoidal variant completes one period every $2\pi/0.1 \approx 63$ steps, which at a sampling intervall of $h = 0.01$ is relatively a rather slow drift than a rapid vibration. Including the noise in our loop, the state transition becomes

$$x_{k+1} = A\, x_k + B\, \tilde{u}_k + n_k^{\text{d}}. \quad (2.33)$$

2.5.5 The full loop

While we, until now, have only regarded each noise type on their own, there is also the possibility to include them all at once. Doing so, a singular step becomes:

$$\begin{aligned}
y_k &= x_k + n_k^{\text{m}}, && (2.34a)\\
u_k &= -K y_k, && (2.34b)\\
\tilde{u}_k &= (1 + n_k^{\text{c}}) \odot u_k, && (2.34c)\\
x_{k+1} &= \left(A + \Delta_k^A\right) x_k + \left(B + \Delta_k^B\right) \tilde{u}_k + n_k^{\text{d}}. && (2.34d)
\end{aligned}$$

Relevant to note about (2.34) is that the controller only interacts with y_k and u_k, whereas the dynamics and the stage cost of (2.19) are evaluated on the true state x_k. Thus our controller can never actually observe the true state it is evaluated

on. For our baseline LQR solution of Section 2.4, which we will use to compare, we compute the gain on a pure system, and only afterwards apply it to an *exposed* system of the style (2.34) afterwards. Measuring how much performance this difference makes is the subject of the next section.

2.6 Metrics and Robustness

In this thesis we try to answer two main questions. First, how well a trained controller performs generally, as well as under influences unknown to him during training. The second relates more to the size of the controller, and a possible tradeoff between performance and the former.

2.6.1 Cost and suboptimality

By Theorem 2.5 no gain achieves a lower cost (2.26) than K^*, and every controller we train is therefore in the best case only as good as it. This serves as our baseline, and we evaluate the cost produces by our controllers in relation to it.

While can easily calculate the infinite-horizon cost of a gain via the Lyapunov equation (2.24), we can't effectively use it. The reason for that is that (2.24) can only be applied to the closed loop, however when adding noise to our evaluation, this is not the case anymore. Additionally, the neural networks we will train later on, don't admit a gain matrix based policy P_K, which would need a different approach anyway.

So, we instead evaluate them over a fixed horizon via rollout. Hence for a given controller, a noise condition c, which is a subset of channels and an intensity Section 2.6.3, and a horizon T, we simulate starting from a randomly drawn $x_0 \sim \mathcal{D}$. The resulting loop (2.34) under c thus accumulates

$$J_T^{(i,s)}(K;c) = \sum_{k=0}^{T-1} \left(x_k^\top Q\, x_k + \tilde{u}_k^\top R\, \tilde{u}_k \right), \tag{2.35}$$

where s indexes the seed the run was trained under and i the episode within it.

Writing $\mathcal{F}$ for the pairs (s,i) whose episode stayed finite, the cost estimate becomes

$$\hat{J}_T(K;c) = \frac{1}{|\mathcal{F}|} \sum_{(s,i)\in\mathcal{F}} J_T^{(i,s)}(K;c), \qquad \mathcal{F} = \left\{ (s,i) \;:\; J_T^{(i,s)}(K;c) < \infty \right\}. \tag{2.36}$$

Our final metric is the relative excess of the cost produced by a trained controller over the cost of the optimal controller K^* under the same condition. It is taken per *per episode*, meaning an episode (s,i) of a trained controller and episode (s,i)

of the baseline are compared here. Both start from the same x_0 and are affected by the same noise. Overall this yields:

$$S^{(i,s)}(K;c) = \frac{J_T^{(i,s)}(K;c) - J_T^{(i,s)}(K^*;c)}{J_T^{(i,s)}(K^*;c)}, \qquad \hat{S}(K;c) = \frac{1}{|\mathcal{F}|} \sum_{(s,i)\in\mathcal{F}} S^{(i,s)}(K;c), \tag{2.37}$$

where $S = 0$ means that the trained controller performed as well as our baseline and $S = 0.1$ that it did ten percent worse.

Note that this metric can indeed be negative. K^* is only optimal for the nominal problem Section 2.4, which has exact state feedback on known matrices and no external disturbance. These conditions do not hold, unless no noise is applied, which is only true if $c = \varnothing$.

2.6.2 Test protocol

For all evaluation throughout this thesis, we will use a standardized approach, so that only the trained controller varies between different runs.

- **Seeds.** $M = 10$ fixed seeds, drawn once randomly.
- **Episodes.** $N = 100$ rollouts per seed, each from an independently sampled $x_0 \sim \mathcal{D}$.
- **Horizon.** $T = 2000$ steps at $h = 0.01$, so a total of 20 seconds.
- **Initial states.** Drawn from the box $[-10, 10] \times [-1, 1]$ of Section 2.4.
- **Initial gain.** For iterative solvers requiring a stable initial gain, we use $K_0 = [\,0.1,\ 0.5\,]$, whose closed loop has $\rho(A - BK_0) \approx 0.9975$.

Under the stronger conditions individual rollouts might diverge. To keep our mean finite, we remove them from the mean calculation, and report them separately as the divergence rate:

$$D(K;c) \coloneqq \frac{1}{MN} \left| \left\{ (s,i) \ : \ J_T^{(i,s)}(K;c) = \infty \right\} \right|. \tag{2.38}$$

An individual episode can be missing from the result due to another reason. Where the *baseline* evaluation of episode (s, i) diverged, $S^{(i,s)}$ has no value at all, since (2.37) would divide by an infinity. Those episodes are excluded from $\hat{S}$ and named seperately.

2.6.3 Noise Intensity

Section 2.5 introduced four channels which change the system compared to the base model. Each noise channel is seperated further into five intensities,

$$\textit{none} \prec \textit{weak} \prec \textit{standard} \prec \textit{strong} \prec \textit{divergent},$$

where *none* leaves the system as it is and thus serves as our control condition. All the other intensities were chosen by applying each channel to the optimal gain K^* and observing its deviation from the optimal path. *divergent*, contrary to the name, doesn't always cause divergence, but makes the controller unusable. The concrete standard deviations per channel are listed in Table A.2.

After an initial analysis of the influence of the different channels separately and combined at their different intensities, we will decide on one of them for the remaining chapters.

2.6.4 The size of a policy

The second question is one of size. For a neural network the known way is simply the number of weights and biases of the actor [CITE: TODO]. For a gain matrix however, we need to find a definition that is comparable to it.

The simply approach for a $K \in \mathbb{R}^{m\times n}$ is the number of its entries, mn. In Chapter 5 however, we will reduce the rank of K. Optimizing the storage size for K of rank r, the singular value decomposition [CITE: SVD_Notes.pdf] seperates it into

$$K = U\Sigma V^\top, \qquad U \in \mathbb{R}^{m\times r}, \quad \Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r), \quad V \in \mathbb{R}^{n\times r}, \tag{2.39}$$

with U and V having orthonormal columns and $\sigma_1 \geq \cdots \geq \sigma_r > 0$ being the non-zero singular values. Hence, we only need to store mr entries for U, r for the diagonal of Σ and rn for V, which in total yields $r(m+n+1)$. For a small enough r this is less, so we take the minimum of both:

Definition 2.6 (Policy size)**.** The size of a linear policy $K \in \mathbb{R}^{m\times n}$ of rank $r = \operatorname{rank}(K)$ is

$$|K| \coloneqq \min\left\{mn,\ r\,(m+n+1)\right\}, \tag{2.40}$$

the number of real numbers required to store it. The size of a neural policy is its total number of trainable parameters in the actor.

Chapter 3

Background

3.1 Motivation

An agent acting in its environment receives observational input, and responds to it with a commanded control input. Differently said, the agent is a function in relation to its environment, and to find an optimal behavior, we need to approximate this optimal function as closely as possible. While Neural Networks are known to be universal approximators [2, 5], and hence able to approximate any function, in the case of an embodied agent, different constraints need to be taken into account. Ay [1] names additionally to external embodiment constraints possible intrinsic constraints, and argues that such can even be "beneficial" for the control structure. Additionally, less parameters for deployment are helpful due to energy usage and storage capacity, the core of the notion of cheap design[CITE: TODO]. So, for such embodied agents, the goal is to find an "embodied universal approximator" [10, p. 3], which restricts the neural network architecture to an intrinsic, unknown dimension, encoding physical properties belonging to the agent and the environment, while disallowing impossible movements [10, Sec. 3].

Biology motivates this approach even further: Here it is clear to see, that by commanding a body, some features of the control architecture can be simplified and delegated to the embodiment[CITE: TODO]. For example, McGeer [8]'s passive walker clearly shows, that some movements can be produced by the body without any control input, just by choosing the right embodiment. While obviously here being simplified to not include any control at all, it can quite simply be extended with such [CITE: #9, reference to his previous paper]. Müller and Hoffmann [11, p. 4] mention such combination, as a robot climbing up walls, whose design was inspired by geckos, indeed using energy for control, however very little compared to an approach not exploiting these specific embodiment design. Further, it can be seen that the complex behaviors are not only less energy intensive to produce, but also utilize muscle synergies [4] and thus fewer control parameters. And while it is difficult to clearly attribute the effect of the embodiment [17], for us, this gives us the reason

to believe that the simplicity, and thus efficiency, comes from a low-dimensionality of the controller [10].

That this is indeed true for certain examples, has been shown both in theory and practice before [3, 9, 7]. Li et al. [7] call this lower-dimension "intrinsic dimension", which is the dimensionality required for the model to be able to find policies. They show for example that for achieving 90 percent of the performance of the full model, the dimensionality can be reduced by 99.6 percent. For a convolutional neural network, it is still 99.4 percent. Others show that the "least complex model which is able to learn an optimal policy does learn faster" [9]. For a neural network reducing parameters can be done in different ways[CITE: TODO], some of which we will take a look at in Chapter 5. In comparison, for LQR, where the controller is a gain matrix, the corresponding approach is reducing its rank [14]. However, while these approaches are more parameter efficient while achieving similar, or slightly worse performance, it was also observed that they tend to perform even worse when encountering noise [3].

Robustness to noise however, is one of the important characteristics of controllers acting under real-world conditions [13, 15]. One approach of achieving this in nature is the principle of "redundancy and diversity" [6], which stands in direct contrast to our goal of reducing redundant parameterizations. Another reason for producing robust controllers is the difficult transfer from digital training environment to real-world embodiments, commonly referred to as the "reality gap" [16, 12]. Approaches of training robust controllers typically introduce noise already during training, be it either by using a LQG or including variations to the dynamics [13, 12]. These shown approaches however, don't look at low-dimensionality, which makes sense as Kitano [6, p. 8] poses that robustness automatically leads to "increased resource demands".

Recent advancements in the research for low-rank policies for LQR...

3.2 Approach

What I will do in the following sections

Chapter 4

Robustness of LQR solvers in different dimensions

4.1 Setting

4.2 Results

4.2.1 Data Sampling Variations for N4SID

Figure 4.1 shows the figure.

Table 4.1 shows the table

Solver	Baseline	N4SID + DARE	Model-based PG + N4SID	S-PI + N4SID	Model-free NPG	S-PI (model-free)
True cost J	$5848 \pm 9.587 \times 10^{-13}$	$5848 \pm 5.175 \times 10^{-5}$	$5848 \pm 2.073 \times 10^{-5}$	5904 ± 0.08024	6105 ± 13.51	7265 ± 47.92
Training suboptimality $(J - J^\star)/J^\star$	0 ± 0	$5.05 \times 10^{-9} \pm 8.85 \times 10^{-9}$	$1.84 \times 10^{-9} \pm 3.55 \times 10^{-9}$	$0.00952 \pm 1.37 \times 10^{-5}$	0.044 ± 0.00231	0.242 ± 0.00819
Test cost†	5968 ± 5272	5968 ± 5272	5968 ± 5272	6025 ± 5322	6233 ± 5510	7422 ± 6531
Suboptimality $S^\dagger$	$+0 \pm 0$	$+5.08 \times 10^{-9} \pm 8.61 \times 10^{-9}$	$+2 \times 10^{-9} \pm 3.85 \times 10^{-9}$	$+0.0116 \pm 0.00734$	$+0.0447 \pm 0.0161$	$+0.318 \pm 0.248$
Environment steps	0 ± 0	500 ± 0	500 ± 0	500 ± 0	$8 \times 10^7 \pm 0$	$8 \times 10^7 \pm 0$
Policy parameters	76 ± 0	76 ± 0	1250 ± 0	76 ± 0	1250 ± 0	965 ± 221
Gain rank	1.0 ± 0.0	1.0 ± 0.0	25.0 ± 0.0	1.0 ± 0.0	25.0 ± 0.0	12.7 ± 2.9

Table 4.1: Solvers learning without the model at the medium inflation, without noise. † mean $\pm$ standard deviation over all 1000 test episodes (100 per seed, 10 seeds) pooled; every other row over the 10 runs.

4.3 Discussion

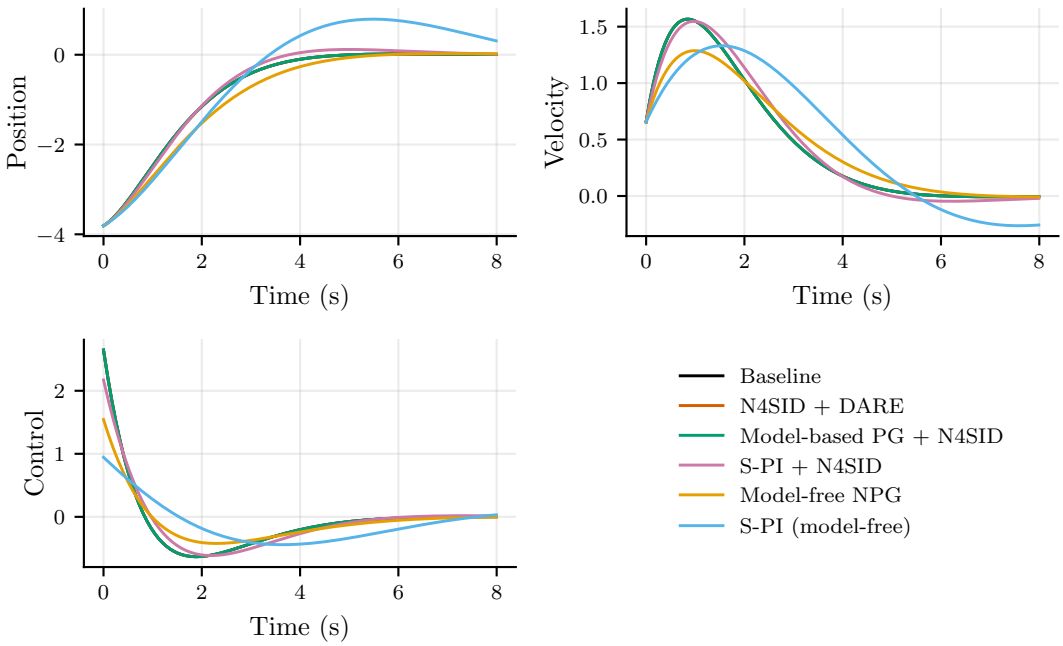


Figure 4.1: TODO

Chapter 5

Robustness under artificially inflated Input

5.1 Setting

5.2 Results

5.3 Discussion

Chapter 6

Perception-based Experiments

6.1 Setting

6.2 Results

6.3 Discussion

Chapter 7

Conclusion

You generally cover three things in the Conclusions section, and each of these usually merits a separate subsection:

1. Conclusions
2. Summary of Contributions
3. Future Research

Conclusions are not a rambling summary of the thesis: they are short, concise statements of the inferences that you have made because of your work. It helps to organize these as short numbered paragraphs, ordered from most to least important. All conclusions should be directly related to the research question stated in the introduction. Examples:

1. The problem stated in the introduction has been solved: as shown in Sections ? to ??, an algorithm capable of handling large-scale Zylon problems in reasonable time has been developed.
2. The principal mechanism needed in the improved Zylon algorithm is the Grooty mechanism.
3. Etc.

The Summary of Contributions will be much sought and carefully read by the examiners. Here you list the contributions of new knowledge that your thesis makes. Of course, the thesis itself must substantiate any claims made here. There is often some overlap with the Conclusions, but that's okay. Concise numbered paragraphs are again best. Organize from most to least important. Examples:

1. Developed a much quicker algorithm for large-scale Zylon problems.
2. Demonstrated the first use of the Grooty mechanism for Zylon calculations.
3. Etc.

The Future Research subsection is included so that researchers picking up this work in future have the benefit of the ideas that you generated while you were working on the project. Again, concise numbered paragraphs are usually best.

Appendix A

Nomenclature

What goes in the appendices? Any material which impedes the smooth development of your presentation, but which is important to justify the results of a thesis. Generally it is material that is of too nitty-gritty a level of detail for inclusion in the main body of the thesis, but which should be available for perusal by the examiners to convince them sufficiently. Examples include program listings, immense tables of data, lengthy mathematical proofs or derivations, etc.

A.1 Notation

The following table contains the conventions regarding symbols used throughout this thesis:

Symbol	**Meaning**
Sets and operators	
I_n, $\mathbf{1}_n$	identity matrix and all-ones vector of dimension n
$\lambda_i(M)$, $\sigma_i(M)$	i-th eigenvalue and i-th singular value of M
$\rho(M)$	spectral radius, $\max_i \lvert\lambda_i(M)\rvert$
$M \succeq 0$, $M \succ 0$	M symmetric positive semi-definite / positive definite
$\lVert\cdot\rVert$, $\lVert\cdot\rVert_F$, $\lVert\cdot\rVert_*$	Euclidean (spectral) norm, Frobenius norm, nuclear norm
$\odot$	elementwise product
$\mathcal{N}(\mu, \Sigma)$	Gaussian distribution with mean μ and covariance Σ
∇f, $\widehat{\nabla} f$	gradient of f and the estimate of it
System description	
n, m, q	dimension of the state, input and output
t, k, h	continuous time, discrete step index, sampling period
$x_k \in \mathbb{R}^n$	state

Symbol	**Meaning**
$u_k \in \mathbb{R}^m$	control input computed by the controller
$\tilde{u}_k \in \mathbb{R}^m$	control input actually applied to the plant
$y_k \in \mathbb{R}^q$	observed output
p, v	position and velocity of the double integrator, $x = [\, p,\ v\,]^\top$
A_c, B_c	continuous-time system and input matrix
A, B	their zero-order-hold discretizations
C, D	output and feedthrough matrix; $D = 0$ throughout
A_K	closed-loop matrix, $A_K = A - BK$
$\mathcal{C}(A, B)$, $\mathcal{O}(C, A)$	controllability and observability matrix
$\hat{x}_k$, $\tilde{x}_k$	state estimate and estimation error $x_k - \hat{x}_k$
L	observer gain
Optimal control	
Q, R	state and input cost matrices, $Q \succeq 0$, $R \succ 0$
$K \in \mathbb{R}^{m \times n}$	feedback gain, $u_k = -Kx_k$
K^*	optimal gain, solution of (2.22)
$\mathcal{K}$	set of stabilizing gains, (2.25)
$J(x_0, u)$	infinite-horizon cost of an input sequence
P	stabilizing solution of the DARE (2.21)
P_K	solution of the Lyapunov equation (2.24) for a fixed K
$\mathcal{D}$, Σ_0	distribution of x_0 and its second moment $\mathbb{E}[x_0 x_0^\top]$
$f(K)$	expected cost of a gain, $\mathrm{tr}(P_K \Sigma_0)$
Noise	
n_k^{m}, σ_{m}	measurement noise and its standard deviation
n_k^{c}, σ_{c}	control noise (relative) and its standard deviation
Δ_k^A, Δ_k^B, σ_{p}	parametric perturbations and their standard deviation
n_k^{d}, σ_{d}	external disturbance and its standard deviation / amplitude
Learning and inflated systems	
d_{in}, d_{out}	inflated input and output dimension, $d \gg n$
U_1, U_2	orthonormal inflation matrices
λ	weight of the nuclear-norm regulariser
η	step size
r	smoothing radius of the zeroth-order gradient estimator
N, T	number of rollouts and rollout horizon

A.2 Noise intensities

The settings used for the different noise intensities. It is important to note that parametric noise is set to be time invariant, so drawn once per rollout, and that the disturbance noise uses its sinusoidal variant (2.32b).

Table A.2: Standard deviation per noise channel and intensity.

Channel	*none*	*weak*	*standard*	*strong*	*divergent*
Measurement σ_{m}	0	0.1	0.5	2.0	20.0
Control σ_{c}	0	0.1	0.5	2.0	10.0
Parametric σ_{p}	0	0.0005	0.001	0.005	0.01
Disturbance σ_{d}	0	0.0005	0.001	0.005	0.05

Appendix B

Additional Proofs

Appendix C

Complete Simulation Results

Step size	3.5e-09	3.5e-08	3.5e-07	3.5e-06	3.5e-05
Stable	10/10	10/10	10/10	10/10	8/10
True cost J	$1.154 \times 10^4 \pm 29.19$	9057 ± 45.26	7517 ± 37.9	6366 ± 30.31	8709 ± 3157
Training suboptimality $(J - J^*)/J^*$	0.974 ± 0.00499	0.549 ± 0.00774	0.285 ± 0.00648	0.0886 ± 0.00518	0.489 ± 0.54
Gain rank	25.0 ± 0.0	25.0 ± 0.0	25.0 ± 0.0	25.0 ± 0.0	24.8 ± 0.7
Spectral radius	0.9975 ± 0.0000	0.9975 ± 0.0000	0.9974 ± 0.0000	0.9959 ± 0.0001	0.9956 ± 0.0030
Test cost†	9528 ± 8601	8237 ± 7406	6804 ± 6071	6085 ± 5429	7753 ± 6593

Table C.1: Model-free S-PI at lambda=400, medium inflation, over 10 seeds. † mean $\pm$ standard deviation over all 200 test episodes (20 per seed, 10 seeds) pooled; every other row over the 10 runs. Stable counts the seeds whose controller left the plant stable.

Step size	3.5e-09	3.5e-08	3.5e-07	3.5e-06	3.5e-05
Stable	10/10	10/10	10/10	10/10	8/10
True cost J	$1.155 \times 10^4 \pm 29.18$	9088 ± 44.71	7834 ± 45.63	7234 ± 77.19	8219 ± 2138
Training suboptimality $(J - J^*)/J^*$	0.974 ± 0.00499	0.554 ± 0.00765	0.34 ± 0.0078	0.237 ± 0.0132	0.405 ± 0.366
Gain rank	25.0 ± 0.0	25.0 ± 0.0	22.9 ± 1.1	10.6 ± 2.3	17.0 ± 8.5
Spectral radius	0.9975 ± 0.0000	0.9976 ± 0.0000	0.9977 ± 0.0000	0.9973 ± 0.0001	0.9970 ± 0.0009
Test cost†	9531 ± 8603	8253 ± 7421	6956 ± 6206	6561 ± 5847	7490 ± 6270

Table C.2: Model-free S-PI at lambda=4000, medium inflation, over 10 seeds. † mean $\pm$ standard deviation over all 200 test episodes (20 per seed, 10 seeds) pooled; every other row over the 10 runs. Stable counts the seeds whose controller left the plant stable.

Step size	3.5e-09	3.5e-08	3.5e-07	3.5e-06	3.5e-05
Stable	10/10	10/10	10/10	10/10	5/10
True cost J	$1.158 \times 10^4 \pm 29.39$	9455 ± 39.97	$3.085 \times 10^4 \pm 3196$	$1.922 \times 10^4 \pm 9017$	$3.482 \times 10^4 \pm 5.301 \times 10^4$
Training suboptimality $(J - J^*)/J^*$	0.981 ± 0.00503	0.617 ± 0.00683	4.28 ± 0.546	2.29 ± 1.54	4.95 ± 9.06
Gain rank	25.0 ± 0.0	17.8 ± 0.6	15.9 ± 3.2	16.4 ± 10.2	8.0 ± 9.2
Spectral radius	0.9975 ± 0.0000	0.9979 ± 0.0000	0.9997 ± 0.0000	0.9988 ± 0.0015	0.9877 ± 0.0139
Test cost†	9552 ± 8623	8433 ± 7584	9726 ± 8707	8323 ± 7830	$3 \times 10^4 \pm 5.527 \times 10^4$

Table C.3: Model-free S-PI at lambda=40000, medium inflation, over 10 seeds. † mean ± standard deviation over all 200 test episodes (20 per seed, 10 seeds) pooled; every other row over the 10 runs. Stable counts the seeds whose controller left the plant stable.

Step size	3.5e-09	3.5e-08	3.5e-07	3.5e-06	3.5e-05
Stable	10/10	10/10	9/10	9/10	0/10
True cost J	$1.205 \times 10^4 \pm 61.54$	$1.331 \times 10^6 \pm 7.803 \times 10^5$	$3.626 \times 10^6 \pm 5.273 \times 10^6$	$3.247 \times 10^4 \pm 2.887 \times 10^4$	–
Training suboptimality $(J - J^*)/J^*$	1.06 ± 0.0105	227 ± 133	619 ± 902	4.55 ± 4.94	–
Gain rank	17.6 ± 0.7	8.4 ± 4.2	4.8 ± 6.0	14.2 ± 9.0	–
Spectral radius	0.9978 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	0.9989 ± 0.0006	–
Test cost†	9784 ± 8839	$1.255 \times 10^4 \pm 1.144 \times 10^4$	$1.233 \times 10^4 \pm 1.135 \times 10^4$	$1.523 \times 10^4 \pm 1.939 \times 10^4$	–

Table C.4: Model-free S-PI at lambda=400000, medium inflation, over 10 seeds. † mean ± standard deviation over all 200 test episodes (20 per seed, 10 seeds) pooled; every other row over the 10 runs. Stable counts the seeds whose controller left the plant stable.

Step size	3.5e-09	3.5e-08	3.5e-07	3.5e-06	3.5e-05
Stable	0/10	0/10	0/10	0/10	0/10
True cost J	–	–	–	–	–
Training suboptimality $(J - J^*)/J^*$	–	–	–	–	–
Gain rank	–	–	–	–	–
Spectral radius	–	–	–	–	–
Test cost	–	–	–	–	–

Table C.5: Model-free S-PI at lambda=4e+06, medium inflation, over 10 seeds. Stable counts the seeds whose controller left the plant stable.

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